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1) Problem Definition

Our group has chosen to do Theme A: Image Processing. We have chosen to implement a three stage processing pipeline, where the output of each stage will feed into the next stage.

1. Grayscale conversion: Convert the RGB values of pixels into an intensity value.
2. Gaussian blur: Apply a 5×5 Gaussian blur to smooth noise and reduce detail prior to the edge detection.
3. Laplacian edge detection: Apply Laplacian edge detection to highlight edges and produce the final edge map image.

2) Data Strategy

Source:

- We will use a set of JPG/PNG images in the public domain and synthetic images as needed for scale and performance testing.

Scalability plan:

- Small: 512×512 for debugging
- Medium: 1920×1080 for standard workload
- Large: 4K or larger for stress testing

3) Goals and Verification Plan

- Goal: Implement serial, OpenMP, and MPI versions of our image processing pipeline.

Verification (correctness):

- To verify the correctness of the image processing, we will compare the parallel outputs generated from OpenMP and MPI against the baseline serial generated output.
- The outputs from each method will be compared against each other to ensure that they match.
- We will implement a comparison step that confirms if the images match or if there are any nonmatching pixels

Performance goals:

- Our performance goal is to achieve near linear speedup with OpenMP before reaching bottlenecks using an appropriate number of threads for the machine we are using
- MPI should have good speedup compared to serial, but with diminishing returns as the latency Increases